

Enhanced Fast-Discharging Performance and Cyclability in Oxygen-Redox-Based P3-Type Na-Layered Cathode via Vacancies in TM layers

Sang-Yeop Lee, Hyunji Kweon, Sangyeop Lee, Min-kyung Cho, Hobin Ahn, Jinho Ahn, Bonyoung Ku, Myungeun Choi, Hun-Gi Jung, Dong Ok Shin,* and Jongsoo Kim*

Oxygen redox in layered oxide cathodes for Na-ion batteries is considered a promising approach for improving the energy density. However, oxygen-redox-based cathodes suffer from sluggish kinetics and undesirable structural change during charge/discharge, leading to poor electrochemical performances. Herein, introducing vacancies ($\square$) in the transition metal layers enables the enhanced oxygen redox-based electrochemical performances in the P3-type Mn-based layered oxide cathode is demonstrated. The vacancies can play a role of the local distortion buffers, resulting in the enhanced oxygen redox kinetics and the suppressed structural deformation such as P3-O3(II) phase transition. The oxygen-redox-based P3-type $\text{Na}_{0.56}[\text{Ni}_{0.1}\text{Mn}_{0.81}\square_{0.09}]\text{O}_2$ exhibits the large discharge capacity of $\approx 140.95 \text{ mAh g}^{-1}$ at 26 mA g^{-1} with a high average discharge voltage of $\approx 3.54 \text{ V}$ (vs Na^+/Na). Even at 650 mA g^{-1} , its discharge capacity and average operation voltages delivered $\approx 122.06 \text{ mAh g}^{-1}$ and $\approx 3.22 \text{ V}$, respectively. Especially, the small gap of average discharge voltage indicates both improves power-capability and enhanced kinetics of oxygen redox in P3-type $\text{Na}_{0.56}[\text{Ni}_{0.1}\text{Mn}_{0.81}\square_{0.09}]\text{O}_2$. Moreover, the vacancy buffer in the transition metal layers results in the stable cycle-performance of P3-type $\text{Na}_{0.56}[\text{Ni}_{0.1}\text{Mn}_{0.81}\square_{0.09}]\text{O}_2$ with the capacity retention of $\approx 80.80\%$ for 100 cycles, due to the suppressed P3-O3(II) phase transition.

vehicles (EVs). Thus, efficient energy storage systems (ESSs) for EVs have garnered significant global attention.^[1–3] In terms of Li-ion batteries (LIBs), they can deliver large energy density and stable electrochemical performances compared to other ESSs. Consequently, the application of LIBs has expanded from the small electronic devices to large-scale ESSs for EVs.^[4–7] However, the limited and concentrated Li sources on Earth pose challenges for ensuring stable production quantities and price stability,^[8–11] which can lead to the highly increased production cost of LIBs in the near future. Therefore, much research have focused on developing low-cost and efficient ESSs based on the earth-abundant sources.

Recently, Na-ion batteries (SIBs) have been considered one of the most promising and low-cost alternatives to LIBs, due to the abundant Na sources in the sea and their similar monovalent-ion-based reaction mechanism like LIBs.^[12–15] In the case of the cathode, layered oxide materials have attracted great attention

1. Introduction

Environmental issues caused by usage of fossil fuels have driven the development of eco-friendly vehicles such as electric

because of the large gravimetric capacities induced by the low molar mass compared to other compounds. Several types of layered oxide cathodes with different Na sites in the structure

S.-Y. Lee, H. Kweon, S. Lee, H. Ahn, J. Ahn, B. Ku, M. Choi, H.-G. Jung, J. Kim
Department of Energy Science
Sungkyunkwan University
Suwon 16419, Republic of Korea
E-mail: jongsookim@skku.edu

S.-Y. Lee, H. Kweon, S. Lee, H. Ahn, J. Ahn, B. Ku, M. Choi, J. Kim
SKKU Institute of Energy Science and Technology (SIEST)
Sungkyunkwan University
Suwon 16419, Republic of Korea
M.-kyung Cho
Advanced Analysis Center
Korea Institute of Science and Technology (KIST)
Seoul 02792, Republic of Korea

The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/aenm.202402412>

DOI: 10.1002/aenm.202402412

H.-G. Jung, J. Kim
Energy Storage Research Center
Korea Institution of Science and Technology
Seoul 02792, Republic of Korea

H.-G. Jung, J. Kim
KIST-SKKU Carbon-Neutral Research Center
Sungkyunkwan University
Suwon 16419, Republic of Korea

D. O. Shin
Smart Materials Research Section
Electronics and Telecommunications Research Institute (ETRI)
Daejeon 34129, Republic of Korea
E-mail: doshin@etri.re.kr

D. O. Shin
Department of Semiconductor and Advanced Device Engineering
University of Science and Technology (UST)
Daejeon 34113, Republic of Korea

have been reported, such as octahedra-site-based (O-type) and prismatic-site-based (P-type) structures. Moreover, it has been reported the P-type layered-oxide cathodes can deliver large available capacities through both the cationic redox by transition metal (TM) cations and the anionic redox by oxygen anions.^[16] In the layered structure, oxygen redox (OR) occurs when oxygen ions in the linear Na-O-X [X: Li, Mg, Zn, *etc.*] arrangement possess unhybridized $2p$ non-bonding orbitals.^[17–19] These orbitals have higher energy levels compared to other hybridized O $2p$ orbitals and can even provide electrons earlier than certain TM t_{2g} orbitals. Due to Mn^{4+} cation with t_{2g} orbitals having lower energy levels than O $2p$ non-bonding orbitals, P-type Mn-based layered oxide cathode can exhibit the OR reaction during Na^+ de/intercalation.^[20,21] However, the OR reaction accompanies undesirable structural deformation in the layered structure due to the presence of oxidized oxygen anions formed in the high-voltage region.^[22] This results in sluggish kinetics of Na^+ diffusion and poor structural stability during prolonged cycling. In terms of P3-type layered oxide cathodes, it was known that they experience the large structural change of P3-O3(II) phase transition during the OR reaction,^[23] and the reported OR-based P3-type layered oxide cathodes just deliver poor electrochemical performances under the SIB system, such as poor cyclability and power-capability. Furthermore, in the OR-based cathodes for SIBs, the operation voltage during discharge is much lower than that during charge, which is even more severe at the fast-discharging process.^[24] Thus, it should be required to suppress the undesirable structural change in the OR-based P3-type layered oxide cathodes during charge/discharge, through introducing the buffer in the structure. It was reported that substitution of vacancies ($\square$) in the TM layer of the layered structure with the triclinic P1 and the hexagonal $P6_3/mmc$ (P2-type) phases results in not only the OR reaction but also the suppressed structural change during charge/discharge.^[25–27] We expected that the presence of vacancies in the TM layer would enable both the highly stabilized OR reaction and the enhanced electrochemical performances of the OR-based P3-type layered oxide cathodes while suppressing undesirable structural deformation.

In this study, we successfully prepared P3-type Mn-based layered oxide cathode with the vacancies, $P3-Na_{0.56}[Ni_{0.1}Mn_{0.81}\square_{0.09}]O_2$. At the current density of 26 mA g^{-1} in the voltage range of $2.2\text{--}4.5\text{ V}$ (vs Na^+/Na), it delivered the specific capacity of $\approx 140.95\text{ mAh g}^{-1}$ corresponding to $\approx 0.55\text{ mol Na}^+$ de/intercalation, which indicates occurrence of both cationic ($Ni^{2+}/^{4+}$) and anionic (O^{2-}/O^{n-} ($n < 2$)) redox reactions. The occurrence of OR reaction in $P3-Na_{0.56}[Ni_{0.1}Mn_{0.81}\square_{0.09}]O_2$ was investigated through ex situ synchrotron-based soft X-ray absorption spectroscopy (sXAS) and first-principles calculation. In particular, the operando X-ray diffraction (XRD) data revealed that even after being charged to 4.5 V (vs Na^+/Na) accompanied by the OR reaction, the presence of vacancies in the TM layers of $P3-Na_{0.56}[Ni_{0.1}Mn_{0.81}\square_{0.09}]O_2$ effectively suppressed the P3-O3(II) phase transition. Through the first-principles calculation, we verified the vacancies can play a role as buffer for suppression of local structural change during the OR reaction. Thus, the discharge capacity of $P3-Na_{0.56}[Ni_{0.1}Mn_{0.81}\square_{0.09}]O_2$ was retained up to $\approx 80.80\%$ of the initial capacity after 100 cycles, indicating the stable cycle-performances. Even at 650 mA g^{-1} , it exhibited the specific capacity of $\approx 122.06\text{ mAh g}^{-1}$, which is

$\approx 86.59\%$ of the capacity measured at 26 mA g^{-1} . Especially, the difference between the average discharge voltages measured at 26 and 650 mA g^{-1} was just $\approx 0.32\text{ V}$, and average voltage gap at 260 mA g^{-1} between the 1st and 100th cycles was negligible, at just 0.0028 V . These results indicate the presence of vacancies in TM layers enables facile and stable OR reaction in the P3-type Na-layered oxide cathode for the improved power-capability and cyclability.

2. Result and Discussion

2.1. Preparation of $P3-Na_{0.56}[Ni_{0.1}Mn_{0.81}\square_{0.09}]O_2$ (P3-NNMVO)

P3-NNMVO was synthesized through the conventional solid-state method. $NaNO_3$, $Ni(NO_3)_2 \cdot 6H_2O$, and $MnCO_3$ were used as precursors, and they were mixed uniformly at 250 rpm for 12 h . After mixing, the obtained mixture was dried at $120\text{ }^\circ\text{C}$ for 12 h and calcined at $590\text{ }^\circ\text{C}$ for 12 h under O_2 gas. **Figure 1a** presents the results of Rietveld refinement based on the XRD pattern of P3-NNMVO. The main Bragg diffraction peaks of P3-NNMVO were indexed to a P3-type layered structure (space group $R3m$) with ABBCCA oxygen stacking, where Na cations occupy the prismatic sites. No impurities or second phases were detected in the XRD pattern. The lattice parameters of P3-NNMVO were calculated as $a = b = 2.86584(2)\text{ \AA}$ and $c = 16.7916(2)\text{ \AA}$, with low reliability factors ($R_p = 1.62\%$, $R_F = 2.09$, and $\chi^2 = 2.18\%$), indicating excellent confidence in the results. To provide more precise structural verification on the vacancies in the TM layer, we conducted neutron diffraction (ND) (**Figure 1b**). The low reliability factors of P3-NNMVO ($R_p = 4.14\%$, $R_F = 7.51$, and $\chi^2 = 6.63\%$) indicate high confidence in the ND-based Rietveld refinement results. Detailed structural information through combined analyses using XRD and ND, including atomic positions, B_{iso} , and occupancies of P3-NNMVO, are summarized in **Table S1** (Supporting Information). Particularly, it was verified that the occupancies of Mn and Ni were designed as $0.8100(9)$ and $0.1000(7)$, respectively, implying presence of $\approx 0.09\text{ mol}$ vacancies in the TM layer.

To provide more direct evidence of the introduced vacancies within the TM layers of P3-NNMVO, the scanning transmission electron microscopy-high-angle annular dark field (STEM-HAADF) analyses were performed. In the STEM-HAADF image (**Figure 1c**), relatively low intensities are irregularly observed. Although Ni in P3-NNMVO has a content similar with vacancies at 0.1 and 0.09 mol , due to its atomic number similarity with Mn, Ni cations exhibit signals resembling those of Mn cations in electron microscopy-based analysis. Consequently, the STEM-HAADF analysis results showed irregularly low intensities within the TM layers, clearly indicating the irregular presence of vacancies in the TM layers of P3-NNMVO. Moreover, the presence of vacancies was confirmed through transmission electron microscopy (TEM) and energy dispersive X-ray spectroscopy (EDS) elemental mapping analyses. As shown in **Figures 1d** and **S1** (Supporting Information), Na, Mn, and Ni elements are homogeneously distributed in the P3-NNMVO particles with an average size of $\approx 250\text{ nm}$. Furthermore, the results from TEM-EDS spectrum analysis for each element (**Figure S2** and **Table S2**, Supporting Information) reveal the Na:Ni:Mn:O ratio of $\approx 0.59:\approx 0.11:\approx 0.80:2.0$, which is well consistent with the elemental ratios in the structural design of P3-NNMVO containing the vacancies in the TM layer

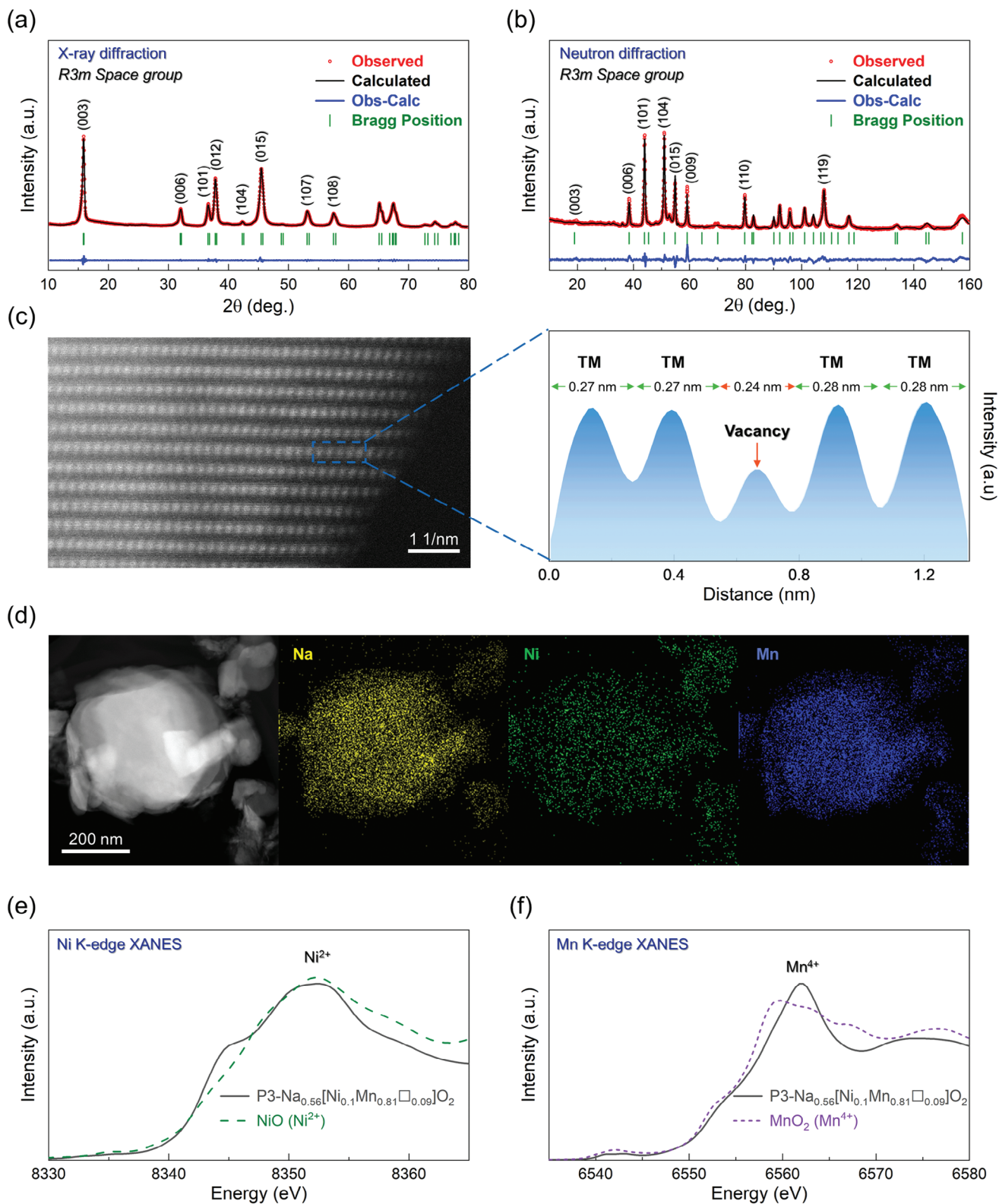


Figure 1. Refined XRD patterns of $\text{P3-Na}_{0.56}[\text{Ni}_{0.1}\text{Mn}_{0.81}\square_{0.09}]\text{O}_2$. b) Refined ND patterns of $\text{P3-Na}_{0.56}[\text{Ni}_{0.1}\text{Mn}_{0.81}\square_{0.09}]\text{O}_2$. c) STEM-HAADF image of $\text{P3-Na}_{0.56}[\text{Ni}_{0.1}\text{Mn}_{0.81}\square_{0.09}]\text{O}_2$ and signal profile of enclosed region by the dotted line. d) STEM image and TEM-EDS mapping results of $\text{P3-Na}_{0.56}[\text{Ni}_{0.1}\text{Mn}_{0.81}\square_{0.09}]\text{O}_2$. Pristine K-edge of e) Ni, and f) Mn.

confirmed through the combined XRD and ND Rietveld analyses. The outcomes of inductively coupled plasma atomic emission spectroscopy (ICP-AES), as presented in Table S3 (Supporting Information), further confirm the atomic ratio of Na, Ni, and Mn in P3-NNMVO as $\approx 0.57:0.1:0.81$, which is well matched with the TEM-EDS results. The TEM-EDS and ICP analyses not only provide supplementary evidence for the structural design, including vacancies, in the XRD and ND structural analyses but also corroborate the introduction of vacancies in the TM layers of P3-NNMVO through compositional analysis alongside the structural analysis. The average oxidation states of Ni and Mn in P3-NNMVO are Ni^{2+} and Mn^{4+} , respectively. X-ray absorption near edge structure (XANES) analysis of Ni in Figure 1e reveals Ni K-edge spectra similar with the NiO spectra, indicating that the average oxidation state of Ni in P3-NNMVO is close to Ni^{2+} . Similarly, XANES analysis of Mn in Figure 1f shows Mn K-edge spectra more closely resembling MnO_2 , suggesting that the average oxidation state of Mn in P3-NNMVO is approaching Mn^{4+} . Additionally, in the transmission electron microscope-electron energy loss spectroscopy (TEM-EELS) analysis, Mn-L_3 and L_2 peaks of P3-NNMVO appeared at ≈ 642.31 eV and ≈ 653.17 eV, respectively, and the ratio of Mn-L_3 peak area to L_2 peak area ($A_{\text{L}_3}/A_{\text{L}_2}$) were ≈ 2.27 , indicating Mn^{4+} oxidation state (Figure S3, Supporting Information).^[28,29]

2.2. Electrochemical Performance of P3-NNMVO with Stable Oxygen Redox

Figures 2a and S4 (Supporting Information) show voltage-capacity profiles of P3-NNMVO measured at the various discharge current densities with constant charge current density of 26 mA g^{-1} in the voltage range of 2.2–4.5 V (vs Na^+/Na).^[30–32] At a current density of 26 mA g^{-1} , its initial charge capacity of $\approx 134.79 \text{ mAh g}^{-1}$ indicates $\approx 0.53 \text{ mol Na}^+$ deintercalation during initial charge via both redox reactions of Ni cations and O anions. Since 0.1 mol of Ni^{2+} ions of P3-NNMVO participate in $\text{Ni}^{2+/4+}$ redox reaction, the cationic redox can contribute to the de/intercalation of 0.2 mol of Na^+ , corresponding to $\approx 50.9 \text{ mAh g}^{-1}$ of capacity. Moreover, as shown in Figure S5 (Supporting Information), it was observed that the slope of the charge curve changes significantly after the capacity of $\approx 50.9 \text{ mAh g}^{-1}$, which implies that the OR reaction in P3-NNMVO has occurred after the $\text{Ni}^{2+/4+}$ redox reaction. The discharge capacity of P3-NNMVO at 26 mA g^{-1} was $\approx 140.95 \text{ mAh g}^{-1}$, indicating the intercalation of $\approx 0.55 \text{ mol}$ of Na^+ ions, with contributions from cationic and anionic redox reactions of $\approx 0.2 \text{ mol}$ and $\approx 0.35 \text{ mol}$, respectively.

Even at the high current density of 650 mA g^{-1} , the discharge capacity was retained to $\approx 122.06 \text{ mAh g}^{-1}$, which maintains $\approx 79.02\%$ of the OR-induced capacity and $\approx 86.59\%$ of the total discharge capacity at 26 mA g^{-1} . These results indicate enhanced power-capability of OR-based P3-NNMVO. Particularly, it was also revealed that the average discharge voltage of OR-based P3-NNMVO was stably retained from $\approx 3.54 \text{ V}$ (vs Na^+/Na) at 26 mA g^{-1} without severe overpotential even as the current density increased. As shown in Figure S6 (Supporting Information), the gap of average voltage between the discharge curves measured at 26 and 650 mA g^{-1} was just $\approx 0.32 \text{ V}$ even during occurrence of OR reaction, which is clearly distinct from the reported OR-based

cathode materials for SIBs that are suffered from the severely enlarged overpotential during discharging at particularly high current densities.^[33,34] In the Ragone plot (Figure 2b), it was also verified that P3-NNMVO exhibited the enhanced power-capability compared to the previously reported OR-based P3-type cathode materials.^[35–37]

To demonstrate the contribution of vacancy to high power-capability of P3-NNMVO, we prepared $\text{P3-Na}_{0.56}[\text{Ni}_{0.1}\text{Mn}_{0.9}]\text{O}_2$ (P3-NNMO) in the same synthesis process. P3-NNMO, unlike other Ni-Mn oxide layered cathodes where OR occurs,^[38–40] exhibits limited attributes for oxygen redox because it contains only 0.1 mol% of Ni. This low Ni content reduced the effect of the reductive coupling effect of Ni-O as minimal, and if all the 3d TM ions, such as Ni and Mn, are oxidized to +4 for OR reaction occur, all the Na^+ ions within the structure must be extracted. Therefore, by comparing P3-NNMVO with P3-NNMO, which does not exhibit oxygen redox, we can effectively demonstrate that P3-NNMVO overcomes the slow kinetics of oxygen redox and improves power-capability through vacancies. Using nudged elastic band (NEB) calculation, we investigated Na^+ diffusion motion and compared required activation barrier energies in $\text{P3-Na}_{0.56}[\text{Ni}_{0.1}\text{Mn}_{0.81}\square_{0.09}]\text{O}_2$ and $\text{P3-Na}_{0.56}[\text{Ni}_{0.1}\text{Mn}_{0.9}]\text{O}_2$, according to the presence of vacancies in the TM layer at the deeply desodiated condition. Figure 2c–d shows the Na^+ diffusion pathways of each material, which are between the Mn layer and Mn-Ni-Vacancy layer in P3-NNMVO, and the Mn layer and Mn-Ni layer in P3-NNMO. It was clearly revealed that the required barrier energy for Na^+ diffusion was lower in P3-NNMVO ($\approx 127.08 \text{ meV}$) than in P3-NNMO ($\approx 377.06 \text{ meV}$). That is, by enhancing charge transfer, vacancies significantly increased the OR kinetics of P3-NNMVO higher than P3-NNMO which has no OR reaction. Furthermore, we proposed that the vacancies are also able to enhance capacity retention of P3-NNMVO during cycling by comparing the cycle-performance of P3-NNMVO and P3-NNMO. As shown in Figure 2e, the discharge capacity of P3-NNMVO was maintained to $\approx 80.80\%$ of the initial capacity after 100 cycles at 260 mA g^{-1} which is better than 77.71% of P3-NNMO even with the higher initial capacity of P3-NNMVO ($\approx 120.78 \text{ mAh g}^{-1}$) than P3-NNMO ($\approx 99.37 \text{ mAh g}^{-1}$). Despite potential issues such as electrolyte side reactions and surface degradation during charging to a high voltage of 4.5 V,^[41] P3-NNMVO exhibits improved cyclability compared to other OR-based P3-type SIB cathodes.^[34,42,43] Moreover, the initial average discharge voltage was $\approx 3.320 \text{ V}$ (vs Na^+/Na) and it was retained to $\approx 3.317 \text{ V}$ after 100 cycles (Figure 2f), which indicates the voltage decay after 100 cycles involving OR reaction was only about 2.81 mV. Furthermore, even at higher current density of 650 mA g^{-1} , P3-NNMVO exhibits enhanced cycle-performance (Figure S7, Supporting Information). The discharge capacity of P3-NNMVO was maintained to $\approx 79.04\%$ of the initial capacity even after 200 cycles at 650 mA g^{-1} which is better than $\approx 48.82\%$ of P3-NNMO with the higher initial capacity of P3-NNMVO ($\approx 106.02 \text{ mAh g}^{-1}$) than P3-NNMO ($\approx 84.33 \text{ mAh g}^{-1}$). Furthermore, the average discharge voltage after 200 cycles at 650 mA g^{-1} for the P3-NNMVO is $\approx 3.265 \text{ V}$, which is only $\approx 5 \text{ mV}$ different from the initial average discharge voltage of $\approx 3.270 \text{ V}$ (Figure S8, Supporting Information). The negligible voltage decay with reduced overpotential of P3-NNMVO during prolonged cycling was also clearly verified in the differential capacity versus voltage (dQ/dV) analyses,

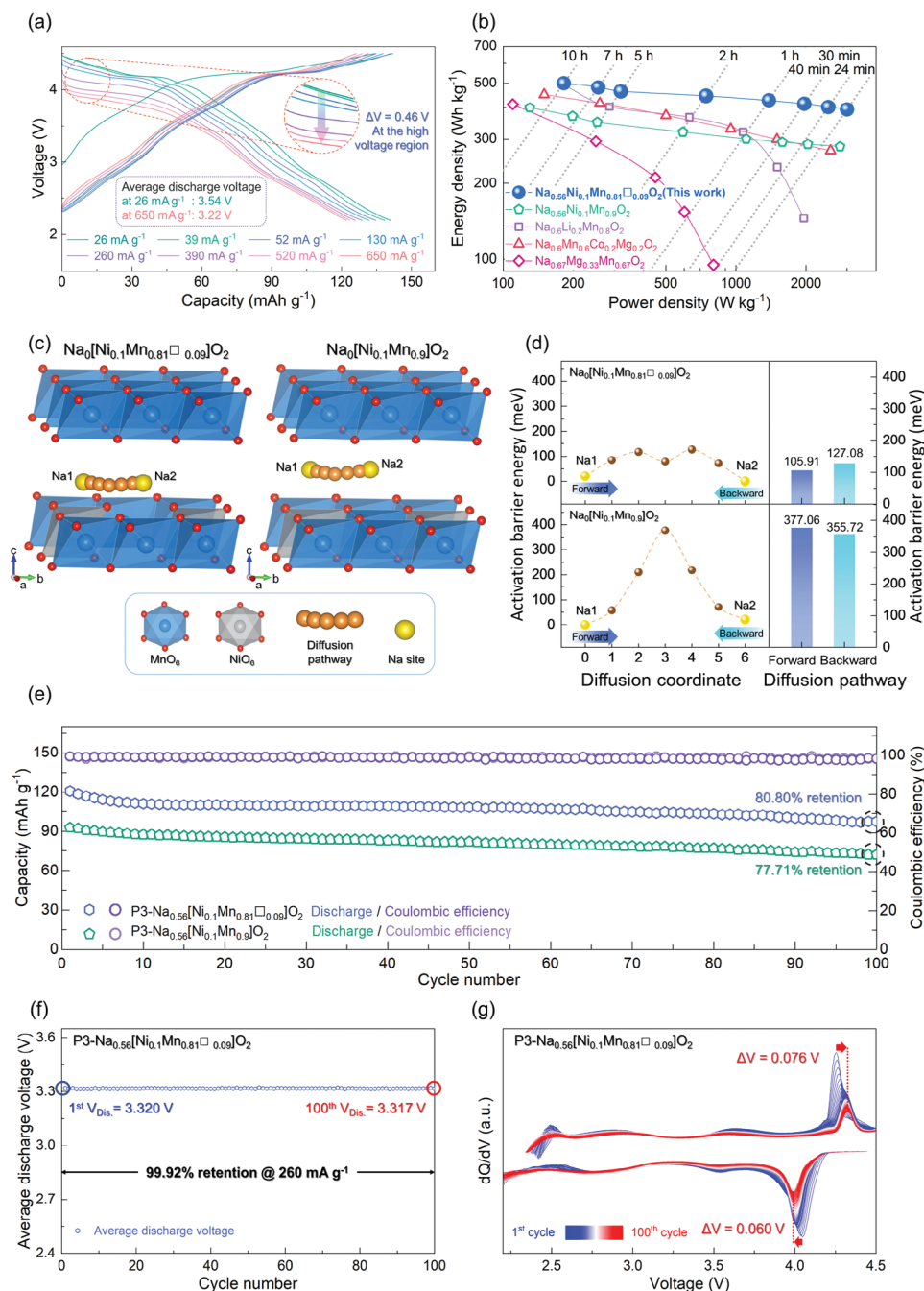


Figure 2. Electrochemical performance of P3-Na_{0.56}[Ni_{0.1}Mn_{0.81}□_{0.09}]O₂: a) Charge/discharge profile of P3-Na_{0.56}[Ni_{0.1}Mn_{0.81}□_{0.09}]O₂ in the voltage range of 2.2–4.5 V (vs Na⁺/Na) at various discharge current densities and a charge current density of 26 mA g⁻¹. b) Comparison of energy and power densities among P3-Na_{0.56}[Ni_{0.1}Mn_{0.81}□_{0.09}]O₂ and other reported Mn-based P3-type layered oxide cathodes for SIBs. c) Predicted Na⁺ diffusion pathway in P3-Na_{0.56}[Ni_{0.1}Mn_{0.81}□_{0.09}]O₂ and P3-Na_{0.56}[Ni_{0.1}Mn_{0.9}]O₂ and d) predicted activation barrier energy for Na⁺ diffusion. e) Cycle performance of P3-Na_{0.56}[Ni_{0.1}Mn_{0.81}□_{0.09}]O₂ and P3-Na_{0.56}[Ni_{0.1}Mn_{0.9}]O₂ in the 2.2–4.5 V (vs Na⁺/Na) voltage range during 100 cycles at 260 mA g⁻¹. f) Average discharge voltages, and g) dQ/dV profiles of P3-Na_{0.56}[Ni_{0.1}Mn_{0.81}□_{0.09}]O₂ during 100 cycles at 260 mA g⁻¹ (in the voltage of 2.2–4.5 V).

contrast to P3-NNMO (Figures 2g and S9, Supporting Information). In the case of P3-NNMVO, the dQ/dV peaks, including those near 4 V corresponding to the oxygen redox reaction, retained their original shapes and positions relatively well even after 200 cycles. However, for P3-NNMO, most of the

dQ/dV peaks observed in the initial cycle underwent the severe shift and lost most of their original shape after 200 cycles. These results indicate that cycle-performance of P3-NNMVO is highly enhanced by the introduction of vacancies in the TM layers. It has been reported that the structure distortion, large

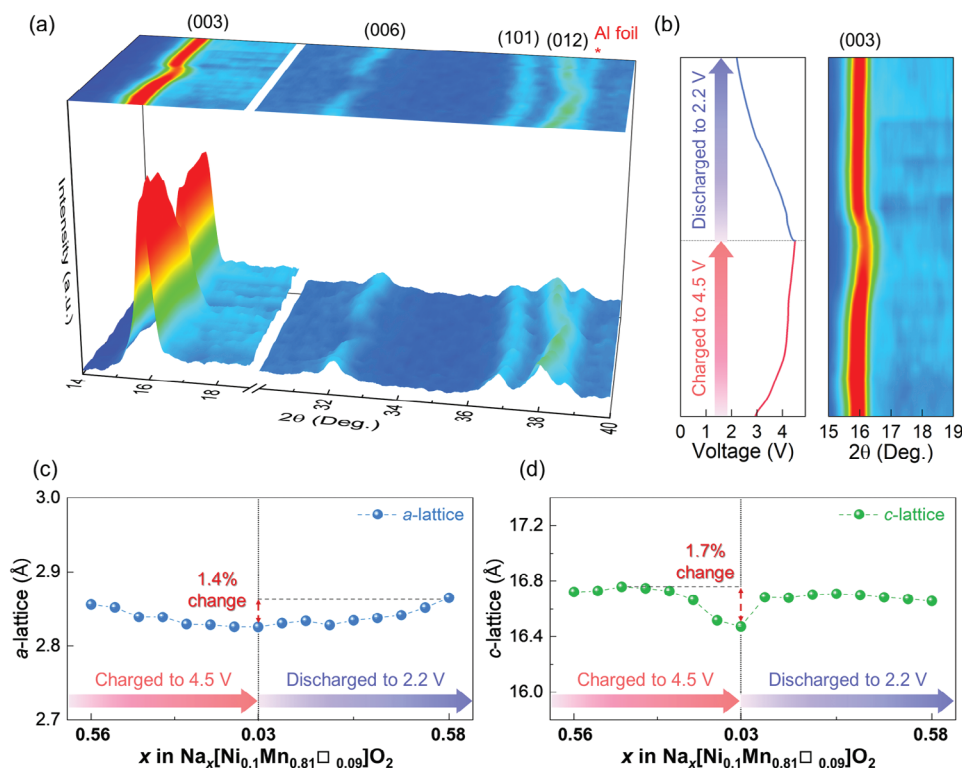


Figure 3. a) Operando XRD pattern of $\text{P3-Na}_{0.56}[\text{Ni}_{0.1}\text{Mn}_{0.81}\square_{0.09}]\text{O}_2$ during charge/discharge. The 2θ angles of collected operando XRD patterns were converted to angles corresponding to $\lambda = 1.54178 \text{ \AA}$ (the wavelength of a conventional X-ray tube source with $\text{Cu K}\alpha$) for easy comparison with other studies. b) Charge/discharge curves and magnified view of (003) planes using results of operando XRD of $\text{P3-Na}_{0.56}[\text{Ni}_{0.1}\text{Mn}_{0.81}\square_{0.09}]\text{O}_2$. c) Change in a , and d) c lattice parameter as a function of Na contents in $\text{P3-Na}_x[\text{Ni}_{0.1}\text{Mn}_{0.81}\square_{0.09}]\text{O}_2$ ($0.03 \leq x \leq 0.58$), verified using Rietveld refinement.

volume change, and the phase transition disturb Na^+ ion migration and sluggish kinetic processes lead to significant overpotential and voltage fade.^[24,44] Furthermore, OR reaction can make structural degradation worse and accelerate the voltage decay during cycling.^[45–47] It implies that suppressing the structural changes during cycling is crucial to improve electrochemical performance, especially for OR-based cathodes. In Figures S10, S11 (Supporting Information), it was revealed that the crystal structure and morphology of P3-NNMVO were well retained even after 100 cycles, and we supposed that due to the role of vacancies as a buffer for volume changes, P3-NNMVO exhibits high structural stability and can prevent the overpotential and voltage drop successfully even after OR.^[48,49]

2.3. Enhanced Structural Durability of P3-NNMVO via Vacancies in the TM Layers

We substantiated the presence of vacancies in the TM layer of P3-NNMVO through structural, compositional, and electrochemical analyses. To confirm effect of vacancies in the TM layer for the structural stabilization, we conducted operando XRD analysis of P3-NNMVO and P3-NNMO during charge/discharge in the voltage window of 2.2–4.5 V (vs Na^+/Na). The complete operando XRD patterns are provided in Figures S12 and S13 (Supporting Information).^[50] As shown in Figure 3a,b, it was verified that the XRD peaks corresponding to (003) plane exhibit monotonic vari-

ations in position and intensity during charge/discharge, without the remarkable behavior of the phase transition. During the initial charging stage, a shift in the direction of lower angles was observed in the (003) peak, indicating an expansion of the interlayer distance in the c -axis direction. The initial c -axis expansion during charging is attributed to the partial extraction of Na^+ ions, leading and enhancing the electrostatic repulsion between O^{2-} ions.^[51,52] As charging progresses, the number of Na^+ ions, which was initially 0.56 mol in the pristine state, decreases to 0.03 mol after charging to 4.5 V. During the charging step in a high-voltage regime, the (003) peak shifts in the direction of higher angles due to the partial reduction in interlayer spacing caused by pronounced Na^+ ion extraction. Particularly, during the charging process to 4.5 V accompanied by the OR reaction, the (003) peak remains consistent, and new XRD peak corresponding to the O3(II)-type phase was negligibly observed. For quantification and comparison on the P3-O3(II) phase transition in P3-NNMVO and P3-NNMO, we analyzed the fraction of O3(II) phase on the full-charged XRD patterns based on the operando XRD analyses by contrasting the intensity ratio of $(003)_{\text{O3(II)}}/(003)_{\text{P3}}$ (Figure S14, Supporting Information). It was verified that the fraction of O3(II) phase in the 4.5 V-charged XRD patterns of P3-NNMVO was just less than 0.1%, which implies that the P3-O3(II) phase transition in P3-NNMVO is negligible. In contrast, in the case of P3-NNMO, the peak intensity of $(003)_{\text{P3}}$ was highly decreased and the new peak at $\approx 18.5^\circ$ (2θ), indicating formation of O3(II) phase, was observed. The fraction

of the O3(II) phase in the 4.5 V-charged P3-NNMO is $\approx 20.19\%$, which is clearly distinct from the results on the 4.5 V-charged P3-NNMVO with vacancies in TM layers.

In most cases of P3-typed layered oxide cathodes, such as P3-NNMO, it was known the electrochemical instability leads to internal distortions that the lattice cannot withstand, causing the TM layers to glide due to electrostatic and cohesive forces.^[35,33] As a result, the interlayer distance decreases, leading to disappearance of the original (003) peak of the P3 phase and emergence of the (003) peak of O3(II) phase at a higher angle. However, in the case of P3-NNMVO, the vacancies in the TM layers serve as the buffer against internal distortions, inhibiting gliding and preventing the transition to P3-O3(II) phase. Moreover, the (101) XRD peak continuously shifted during charge/discharge without formation of a new XRD peak.

To investigate the structural changes occurred during charge/discharge in detail, we investigated the changes on lattice parameters of P3-NNMVO during charge/discharge through Rietveld refinement on the operando XRD patterns. As shown in Figure 3c,d, during the charging process to 4.5 V accompanied by the OR reaction, the variation in the *a*- and *c*-lattice parameters were just $\approx 1.0\%$ decrease and $\approx 1.7\%$ decrease, respectively. These changes are smaller than those observed in P3-type layered cathodes undergoing the known P3-O3(II) phase transition. Furthermore, the *a*- and *c*-lattice parameters were almost restored to the original states during subsequent discharging to 2.2 V. The total changes of *a*- and *c*-lattice parameters during charge/discharge were just $\approx 1.4\%$ and $\approx 1.7\%$, respectively. Whereas the original P3-type phase of P3-NNMVO was well retained during charge/discharge, it was observed that P3-NNMO experiences large structural changes resulting in the phase transition of P3-O3(II) (Figure S15, Supporting Information). After charging to 4.5 V (vs Na⁺/Na), the peak intensity corresponding to the (003) plane was highly decreased and the new peak at $\approx 18.5^\circ$ (2θ), indicating formation of O3(II) phase, was clearly observed,^[35] which is different from P3-NNMVO with the negligible phase transition. These results indicate that the vacancies in the TM layer of P3-NNMVO successfully play a role as the buffer to prevent the severe structural change during charge/discharge containing the OR reaction.

In addition, the first-principles calculation was performed to confirm the enhanced structural stability in P3-NNMVO via the vacancies in the TM layer in detail. As shown in Figure 4a,b, we conducted a comparison of the theoretical changes in the local environments of MnO₆ octahedra between P3-Na_x[Ni_{0.1}Mn_{0.81}□_{0.09}]O₂ (P3-Na_xNMVO) and P3-Na_x[Ni_{0.1}Mn_{0.9}]O₂ (P3-Na_xNMO) during Na⁺ de/intercalation. It was verified that Mn-O bonding in P3-Na_xNMVO was more stably retained during 0.67 mol Na⁺ de/intercalation than that in P3-Na_xNMO, although OR is occurred in P3-Na_xNMVO unlike P3-Na_xNMO. Interestingly, it was verified that with suppressed local environmental changes of MnO₆, the vacancy size was also changed during Na⁺ de/intercalation. We also predicted the change of the vacancy area in three compositions of P3-Na_xNMVO (*x* = 0.67, 0.33 and 0). The area of the vacancy was determined by the area of a triangle, calculated with three surrounding oxygen atoms as reference points. As shown in Figure 4c, it was verified that the area of vacancy was changed depending on the Na contents in the P3-Na_xNMVO structure, implying that the

vacancies in the TM layers can act as a buffer to suppress the large structural change occurring during Na⁺ de/intercalation. Additionally, we investigated the structure when the Na content in P3-Na_xNMVO exceeds its pristine state, which can occur during discharging. As depicted in Figure S16 (Supporting Information), the theoretical model of P3-Na_{0.67}NMVO was slightly tilted compared to the ideal P3-layered structure but still maintained the P3-layered structure.^[54] In particular, the theoretical model of P3-Na_{0.67}NMVO was well matched with ex situ XRD results of the P3-Na_{0.67}NMVO via discharging to ≈ 2.09 V (vs Na⁺/Na) (Figure S17, Supporting Information). Therefore, we were able to determine that the P3-layered structure was well preserved during not only the discharge process but also the charge process.

2.4. OR-Based Reaction Mechanism of P3-NNMVO

We compared *p*DOS when *x* = 0.33, 0 to confirm the redox mechanism in the deeply desodiated area (Figure S18, Supporting Information). In the case of P3-Na_{0.33}NMVO, it was verified that the electron density of O 2*p* orbital near Fermi level (*E*_F) is larger than those of Ni and Mn 3*d* (TM 3*d*) orbitals, which implies that the electrons for further Na⁺ deintercalation from P3-Na_{0.33}NMVO can be provided by O 2*p* orbital rather than TM 3*d* orbitals. Particularly, in the hole densities of P3-Na₀NMVO near *E*_F, the O 2*p* orbital has larger DOS intensity than TM 3*d* orbitals, indicating the Na⁺ deintercalation from P3-Na_{0.33}NMVO to P3-Na₀NMVO can be occurred through oxygen redox reaction. On the other hand, in the case of P3-Na_{0.33}NMO with no vacancies, the electron density of O 2*p* and TM 3*d* orbitals near *E*_F are similar with each other. Moreover, after all Na⁺ deintercalated from P3-Na_{0.33}NMO, the hole density of Ni 3*d* orbital near *E*_F is larger than that of O 2*p* orbital, which mean that the Na⁺ deintercalation from P3-Na_{0.33}NMO to P3-Na₀NMO are majorly determined by the redox reaction of Ni ions rather than oxygen redox reaction. Also, considering the band gap in P3-Na₀NMVO was 0.91 eV narrower than in P3-Na₀NMO, we could expect much easier electronic conduction in P3-Na_xNMVO than in P3-Na_xNMO, in the high-voltage region where the oxygen redox reaction occurs.^[11] Moreover, using DFT calculation, it was also clearly revealed that the activation barrier energy for Na⁺ diffusion can be lowered through introduction of vacancies in TM layers (Figure 2c,d). When the crystal structures of the cathode materials are transformed into different phases or the structure distortion increases, Na⁺ ions are getting difficult to migrate and sluggish kinetic processes lead to extreme polarization and large voltage hysteresis.^[24,44] In operando XRD results (Figure 3), it was verified that P3-NNMVO with just exhibits the small changes of the lattice parameters and negligible phase transition during charge/discharge. It implies that the voltage hysteresis can be improved by suppressing phase transition and restraining internal structural deformation during charge/discharge, due to introduction of vacancies in the TM layers. These results indicate the enhanced ionic and electronic conductivity of P3-NNMVO through introduction of vacancies in TM layers, enabling significant improvement in voltage hysteresis even during fast-discharging process.

To clarify the redox reaction of O anions neighboring the vacancy, we visualized variation in the electron and hole densities

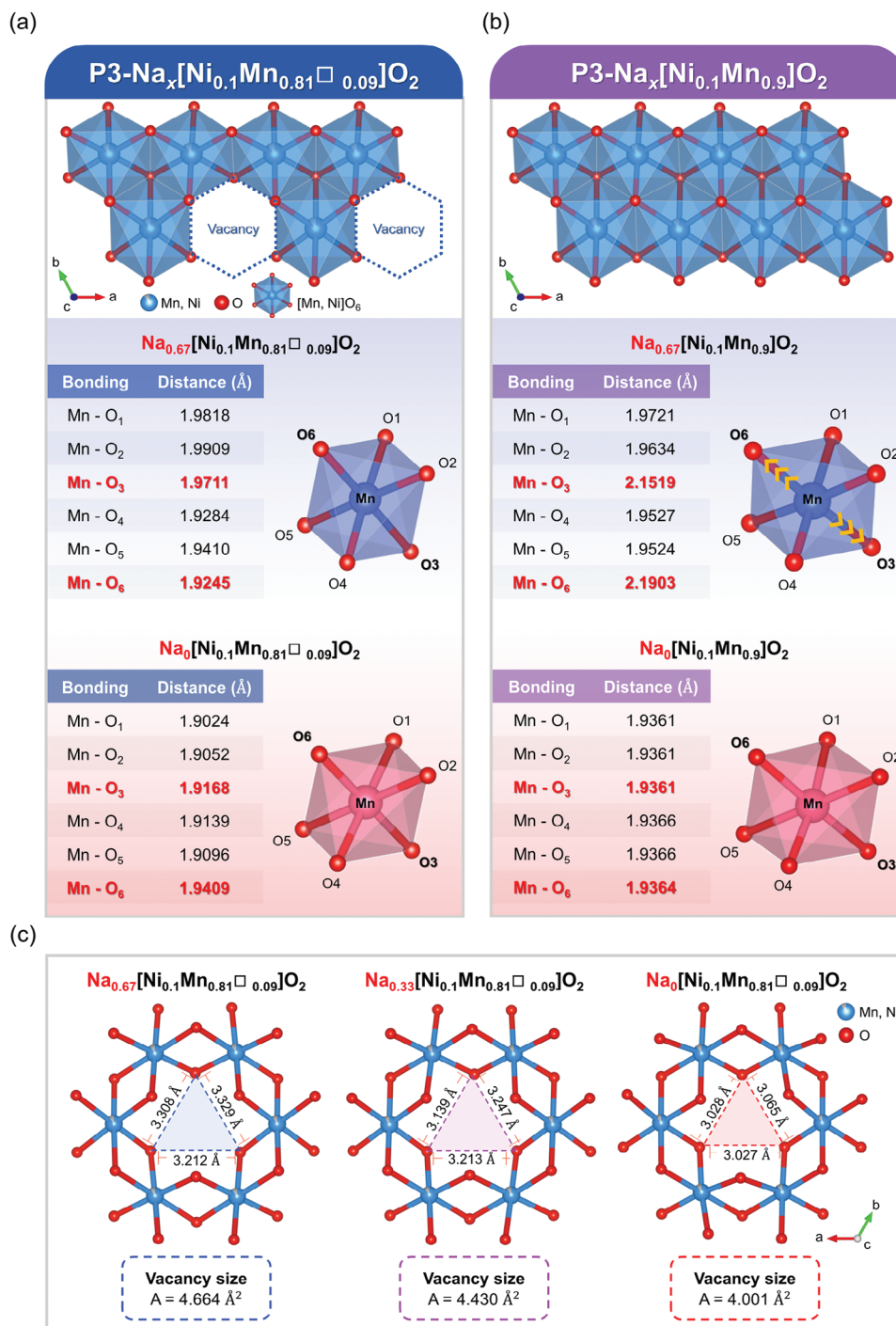


Figure 4. Comparison of predicted Mn–O bond distance of MnO₆ octahedra between a). P3-Na_x[Ni_{0.1}Mn_{0.81}□_{0.09}]O₂ and b) P3-Na_x[Ni_{0.1}Mn_{0.9}]O₂ ($x = 0.67$, and 0). c) Vacancy size prediction of P3-Na_x[Ni_{0.1}Mn_{0.81}□_{0.09}]O₂ ($x = 0.67$, 0.33, and 0).

near E_F of the P3-Na_xNMVO from the DOS results. The yellow region corresponds to the electron orbitals, and the blue region corresponds to the hole orbitals (Figure 5a). As the Na contents in the structure was decreased from 0.33 mol to 0 mol, a significant increase in the pDOS of O 2p orbital was observed at near E_F compared to those of other TM 3d orbital. It implies that OR is more dominant than TM-based cationic redox during Na⁺ deintercalation.

tion between P3-Na_{0.33}NMVO and P3-Na₀NMVO. To clarify the redox reaction of O anions neighboring the vacancy, we visualized variation in the electron and hole densities near E_F along the c -axis direction. As shown in Figure 5b, the electron densities of O 2p orbital in P3-Na_{0.33}NMVO was changed to the hole density during Na⁺ deintercalation in stages, indicating occurrence of OR in P3-NMVO during charge/discharge. The redox

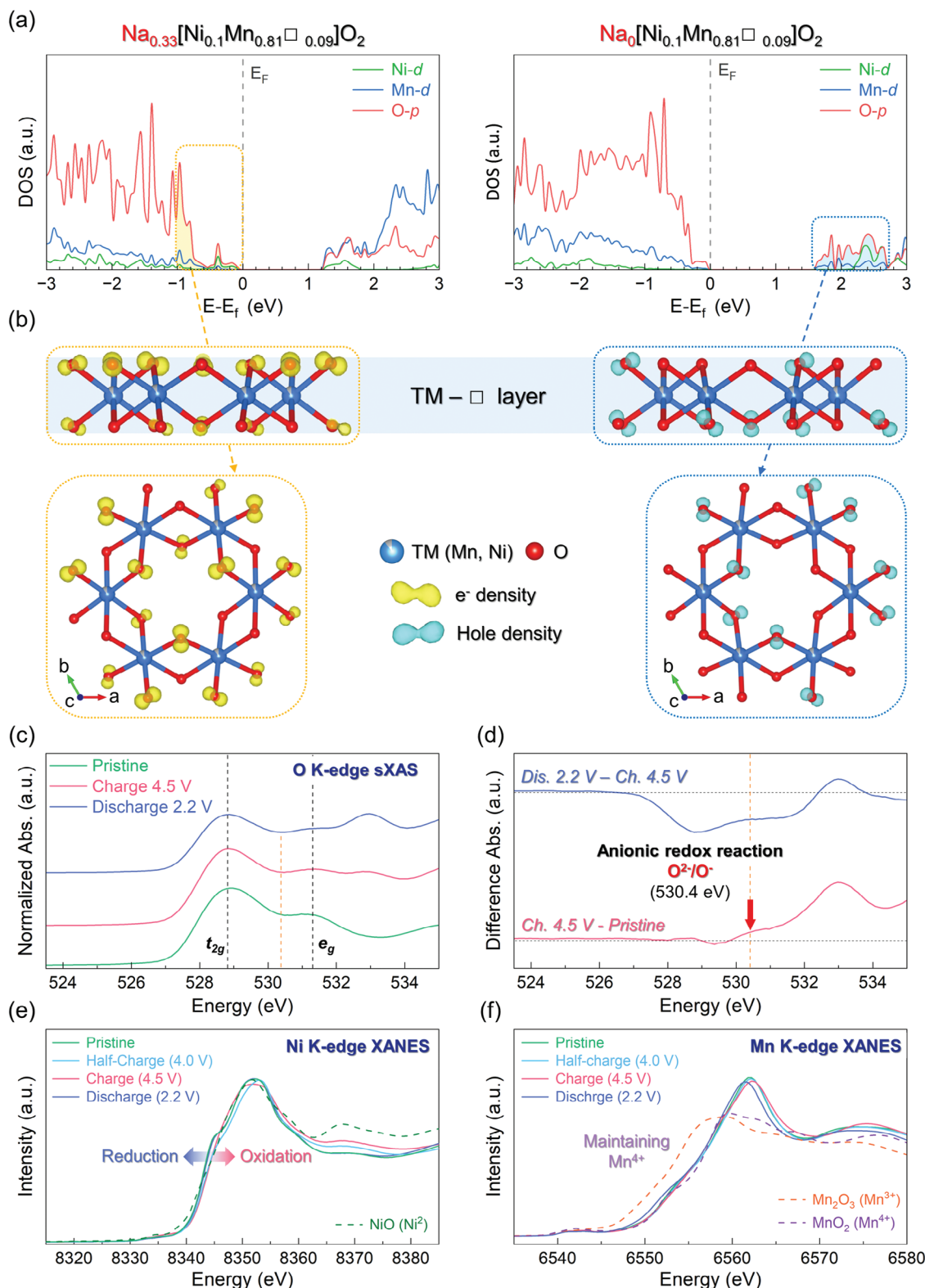


Figure 5. a) pDOS of Ni-d, Mn-d, and O-p orbitals in P3-Na_x[Ni_{0.1}Mn_{0.81}□_{0.09}]O₂ (x = 0.33, and 0). b) Visualized pDOS of electron/hole density in P3-Na_x[Ni_{0.1}Mn_{0.81}□_{0.09}]O₂ (x = 0.33, and 0). c) O K-edge sXAS spectra and d) Difference spectra on O K-edge of P3-Na_{0.56}[Ni_{0.1}Mn_{0.81}□_{0.09}]O₂. e) Ni K-edge XANES spectra and f) Mn K-edge XANES spectra during charge/discharge.

reactions of TM cations and O in P3-NNMVO were also confirmed through the prediction on the integrated spin moments of each element. As shown in Figure S19a (Supporting Information), the magnetic moment (μ_B) of Mn cation remained relatively constant during Na^+ deintercalation from P3- $\text{Na}_{0.5}$ NMVO and P3- Na_0 NMVO, indicating that Mn cation does not participate in the redox reaction during Na^+ deintercalation from P3- $\text{Na}_{0.5}$ NMVO and P3- Na_0 NMVO. On the other hand, in the case of O anion (Figure S19b, Supporting Information), the μ_B was maintained during Na^+ deintercalation from 0.5 mol to 0.33 mol and then, it was changed after further Na^+ deintercalation from 0.33 mol to 0 mol. These calculation results imply that OR has occurred in P3-NNMVO during Na^+ deintercalation. Based on these results, we could expect that the formation of the $\text{Na-O-}\square$ configuration within P3- Na_x NMVO would result in the O 2p non-bonding orbital at a high energy level near the E_F , and thus the oxygen redox reaction would be more easily induced.^[55,56]

To confirm the occurrence of the OR reaction ($\text{O}^{2-}/\text{O}^{n-}$ ($n < 2$)) in P3-NNMVO, we conducted *ex-situ* O K-edge soft X-ray absorption spectroscopy (sXAS) analyses with the total fluorescence yield (TFY) mode. In Figure 5c, the pre-edge peaks of the O K-edge indicate the unoccupied O 2p orbital hybridized with the TM 3d orbital ($\text{TM}_{3d}\text{-O}_{2p}$), which are distinguished by peaks at ≈ 529 and ≈ 531.3 eV, each corresponding to the hybridization with the t_{2g} and e_g orbitals of TM cations. The strong hybridization between the O 2p orbital and the TM 3d orbital, which facilitates electrons to easily jump from the O 2p orbital to the transition metal 3d orbital, can promote oxygen ions to participate in redox reactions and makes the hole states in $\text{TM}_{3d}\text{-O}_{2p}$ hybridized orbital, leading to an increase in the intensities of the peaks at ≈ 529 and ≈ 531.3 eV.^[57,58] Figure S20 (Supporting Information) indicates the variations of the pre-edge integrated intensity during the charge/discharge of P3-NNMVO and significantly increased intensity at charge state than pristine implies the oxygen ions participate in redox reactions. Furthermore, the increase in intensity at ≈ 530.4 eV was reported to be attributed to oxidized oxygen ions.^[59,60] As shown in Figure 5d, the increase in intensity at ≈ 530.4 eV during charging up to 4.5 V provides evidence for the occurrence of OR in P3-NNMVO.^[61,62] Additionally, the peak at ≈ 533 eV is considered to be associated with carbonate functional groups resulting from electrolyte side reactions occurring at the surface.^[63,64] To demonstrate the reversibility of the OR caused by vacancies in P3-NNMVO, an operando differential electrochemical mass spectrometry (DEMS) analysis was performed to determine the oxygen evolution during the initial cycle. (Figure S21, Supporting Information). The operando DEMS analysis showed that there is no O_2 gas releasing from the P3-NNMVO even during charging to 4.5 V (vs Na^+/Na), which indicates the stabilization and enhanced reversibility of oxygen redox reaction in P3-NNMVO via introduction of vacancies in the TM layers.

The cationic redox reactions of Mn and Ni cations in P3-NNMVO were investigated using *ex situ* X-ray absorption near-edge structure (XANES) analyses. Figure 5e shows the Ni K-edge was shifted toward higher energy level during charging to 4.0 V, indicating oxidation of Ni^{2+} cations.^[65] When subsequently charged to 4.5 V, the shape of the Ni spectrum is slightly changed. This phenomenon is attributed to the oxidation of oxygen ions near Ni, and it can also be an indicator that oxygen participated in the redox reaction.^[43] Upon discharging to 2.2 V, the Ni K-

edge was re-shifted to lower energy level similar to the pristine state (Ni^{2+}). In the case of Mn cations in P3-NNMVO, the K-edge was negligibly shifted during charging to 4.5 V and the shape was changed, indicating both retention of Mn^{4+} oxidation state during charge and occurrence of OR (Figure 5f). And, upon discharging, the Mn K-edge was slightly shifted toward lower energy level, indicating a few Mn^{4+} cations are reduced to Mn^{3+} . Additionally, in the TEM-EELS analysis, the ratio of Mn- L_3 peak area to L_2 peak area (A_{L3}/A_{L2}) were ≈ 2.27 and 2.37 respectively in pristine state and charged state, indicating Mn^{4+} oxidation state for both of them (Figure S22, Supporting Information).^[28] The extended x-ray absorption fine structure (EXAFS) analysis reveals the elongation/shrinkage of Ni-O bonds^[76] (D. C. Koningsberger, 1988) (Figure S23, Supporting Information), enabling the observation of local structural changes in P3-NNMVO during charge/discharge. During the charging process, the Ni-O bonding distance decreases from 1.50 to 1.45 Å due to the oxidation of Ni^{2+} . After discharge, the distance almost returns to its pristine state, and the peak shapes in the EXAFS spectra also remain consistent throughout the charge/discharge. Moreover, in the case of Mn-O bonding interaction, it was reversibly changed during charge/discharge, because of local structural change by redox reaction of Ni and O ions, while the peak at ≈ 0.37 Å was generated by data fitting process^[66] and does not represent any significant information about the structure.^[67] After discharging, the M-O bonding intercalation was almost recovered to the pristine state (Figure S24, Supporting Information).

These results indicate that P3-NNMVO undergoes reversible structural changes during charge/discharge accompanied by the OR reaction, which is well matched with the operando XRD and calculation results.

3. Conclusion

The enhanced cyclability of P3-NNMVO is attributed to the small structural change during charge/discharge. Through the detailed structural analyses, we confirmed that vacancies in the TM layer can successfully suppress the large structural change during charge/discharge accompanied by the OR reaction. First-principles calculations also suggest a mechanism by which vacancies can act as a buffer against internal distortions during the charge/discharge process, thereby improving structural stability. It was clearly demonstrated the presence and amount of vacancies introduced into the TM layer of the P3-type layered structure SIB cathode, and propose a mechanism for its performance enhancement. In particular, our research moves beyond simply stabilizing the OR reaction through the introduction of vacancies. Our approach involves introducing a combination of Ni-Mn-Vacancy in the TM layers, which significantly enhances not only the stabilization of the OR reaction but also improves the slow kinetics of the OR reaction. This enables us to minimize overpotential during the fast-discharging process, achieve the high energy density, and attain outstanding power capability. These aspects clearly differentiate our work from previous studies. We believe that our research presents the easy way to enhance the poor fast-discharging performance and cyclability of OR-based cathode materials for high-energy and high-power Na-ion batteries. The discharge capacity and the average discharge voltage of P3-NNMVO, which were ≈ 140.95 mAh g^{-1} and ≈ 3.54 V (vs Na^+/Na)

at 26 mA g⁻¹, respectively, were retained to ≈122.06 mAh g⁻¹ and ≈3.22 V at 650 mA g⁻¹. Notably, the gap between the average voltages measured at 26 and 650 mA g⁻¹ was just 0.32 V, indicating the facile kinetics of OR in P3-NNMVO. Moreover, for 100 cycles at 260 mA g⁻¹, the discharge capacity of P3-NNMVO was maintained to ≈80.80% of the initial capacity, with the high Coulombic efficiency (>99%), surpassing the cycle-performance of P3-NNMO with no vacancy in the TM layer. In the case of operando XRD analyses, it was revealed that the XRD peaks of P3-NNMVO are monotonously changed during charge/discharge with suppression of undesirable P3-O3(II) phase transition, and the variation of the *a*-, *c*-lattice parameters were just ≈1.4% and ≈1.7%, respectively. Furthermore, first-principles calculation and EXAFS results indicate P3-NNMVO experience small and reversible local structural change with suppressed distortion of TMO₆ octahedra during Na⁺ de/intercalation. We believe that our research presents the easy way to enhance the poor fast-discharging performance and cyclability of OR-based cathode materials for high-energy and high-power Na-ion batteries.

4. Experimental Section

Synthesis of P3-Na_{0.56}(Ni_{0.1}Mn_{0.81}□_{0.09})O₂: P3-Na_{0.56}(Ni_{0.1}Mn_{0.81}□_{0.09})O₂ compound was synthesized through the conventional solid-state method. NaNO₃ (Daejung, 99.0%), Ni(NO₃)₂·6H₂O (Alfa, 99%), and MnCO₃ (Alfa, 99.9%) were used as precursors and excess NaNO₃ was added to compensate for the loss of Na at high temperatures. The total weight of 10 g of the precursors, 100 ml of ethanol (Samchun, 99%), and 25 zirconia ball (Ø 10 mm) was sealed in a nalgene low density polyethylene (LDPE) bottle, and they were mixed uniformly using ball-milling machine (Wisd, BML-2) at 250 rpm for 12 h. After mixing, the obtained mixture was dried at 120 °C in the forced conventional oven (JEIO TECH, OF 11E) for 12 h. The dried mixture was pelletized under 300 bar and calcined at 590 °C (5 °C min⁻¹ of heating rates) for 12 h under 0.3 L min⁻¹ flow of O₂ gas and then slowly cooled down to 200 °C in the same O₂ condition without limiting the cooling rate. After cooling, the powder was immediately moved to the Ar-filled glovebox to avoid the moisture contact in the air condition.

X-ray Diffraction: The crystal structure of P3-Na_{0.56}(Ni_{0.1}Mn_{0.81}□_{0.09})O₂ was analyzed using X-ray diffraction (XRD, PANalytical Empyrean) with Mo K_α radiation (λ = 0.70932 Å) in the 2θ range of 4.61°–34.32° with a step size of 0.01°. Then, the angles of the XRD patterns were converted with Cu K_α radiation (λ = 1.54178 Å) for easy comparison with other studies. Rietveld refinement was conducted using the FullProf software based on the obtained XRD data. Operando XRD (o-XRD) patterns of P3-Na_{0.56}(Ni_{0.1}Mn_{0.81}□_{0.09})O₂ were obtained during charge/discharge at a current density of 26 mA g⁻¹ within the voltage range of 2.2–4.5 V (vs Na⁺/Na).

Neutron Diffraction: The crystal structure of P3-Na_{0.56}(Ni_{0.1}Mn_{0.81}□_{0.09})O₂ was also analyzed using Neutron diffraction conducted from the HANARO facility at the Korea Atomic Energy Research Institute (KAERI). The 2 θ range was 0°–160° with a step size of 0.05°, and the neutron wavelength was 1.83458 Å.

Inductively Coupled Plasma-Atomic Emission Spectroscopy: The atomic ratios of Na, Ni, and Mn were determined using an inductively coupled plasma emission spectrometer (ICP-AES; OPTIMA 8300, Perkin-Elmer) at the National Center for Inter-University Research Facilities (NCIRF) at Seoul National University.

Scanning Electron Microscopy: To investigate the particle morphology and microstructure of P3-Na_{0.56}(Ni_{0.1}Mn_{0.81}□_{0.09})O₂, field-emission scanning electron microscopy (FE-SEM; Gemini SEM 560, ZEISS) was performed at the National Center for Inter-University Research Facilities (NCIRF) at Seoul National University.

Transmission Electron Microscopy-Energy-Dispersive X-ray Spectroscopy: To determine atomic ratios and particle morphology, Transmission elec-

tron microscopy-energy-dispersive X-ray spectroscopy (TEM-EDS; T 1071, Elite) and Transmission electron microscopy (TEM; ARM 200F, LEM) was performed at the Cooperative Center for Research Facilities (CCRF) at Sungkyunkwan University.

Scanning Transmission Electron Microscopy-High-Angle Annular Dark Field: The presence of vacancies in P3-Na_{0.56}(Ni_{0.1}Mn_{0.81}□_{0.09})O₂ was confirmed using scanning transmission electron microscopy-high-angle annular dark field (STEM-HAADF) analysis at the Korea Institute of Science and Technology (KIST).

High-Resolution Transmission Electron Microscopy-Electron Energy Loss Spectra: The valence states of Mn in the P3-Na_{0.56}(Ni_{0.1}Mn_{0.81}□_{0.09})O₂ were determined using high-resolution transmission electron microscopy (HR-TEM; FEI TITAN 80–300) at the Korea Institute of Science and Technology (KIST). Electron energy loss spectra (EELS) were collected using a Gatan Quantum 966 spectrometer.

X-ray Absorption Spectroscopy: The valence states of Ni and Mn in the P3-Na_{0.56}(Ni_{0.1}Mn_{0.81}□_{0.09})O₂ were determined using ex situ X-ray absorption spectroscopy (XAS) spectra at the 7D beamlines at the Pohang Accelerator Laboratory (PAL); Cu and Mn metal foils were used as reference spectra. The Ni and Mn K-edge spectra were collected in transmission mode with an electron energy of 2.5 GeV and 200 mA of the current condition. The normalized XAS spectra were calibrated using linear regression to fit the pre-edge region and a quadratic polynomial to the post-edge region in the Athena program, using each metal foil spectrum. To confirm the valence states of O, the total fluorescence yield (TFY) mode was measured using soft X-ray absorption spectroscopy (sXAS) spectra with high-energy grating (HEG) at the 4D PES beamline at PAL with a 525–560-eV photon range.

Differential Electron Mass Spectrometry: The quantities of O₂ gas produced throughout the electrochemical examination was identified using differential electron mass spectrometry (DEMS; Hidden Analytical, UK). Coin cells featuring an aperture in the casing were utilized for the DEMS test and placed inside the DEMS cell to facilitate the inflow and outflow of gases. The cell was connected to the DEMS analysis line, which consisted of an argon (Ar) carrier gas, a mass flow controller, an inlet/outlet gas line for cell connection, and a mass spectrometer. Ar gas (99.999% purity) was employed as the carrier gas, and the gas flow rate was adjusted to 2.5 mL min⁻¹. The internal area of the cell was purged with Ar gas to eliminate any residual impurities. Following the purging process, the cell underwent charging/discharging at a current density of 26 mA g⁻¹ within the voltage range of 2.2–4.5 V (vs Na⁺/Na).

Electrochemical Characterization: An P3-Na_{0.56}(Ni_{0.1}Mn_{0.81}□_{0.09})O₂ electrode was generated from a slurry of 80 wt% active material, 10 wt% Super P as the conducting carbon, and 10 wt% poly(vinylidene fluoride) (PVDF) as a binder in N-methyl-2-pyrrolidone (NMP). The mixed slurry was applied onto Al foil and dried at 100 °C for 12 h in a vacuum oven. The dried electrode was electrochemically tested using CR2032-type coin cells assembled as half-cells, using Na metal as a reference/counter electrode, a separator (GF/F glass fiber filter), and an electrolyte (1 M NaPF₆ in propylene carbonate (PC) and fluoroethylene carbonate (FEC) in a volume ratio of 98:2) in an Ar-filled glove box in the SIB system. Galvanostatic charge/discharge tests were performed at various current densities (26, 39, 52, 130, 260, 390, 520, and 650 mA g⁻¹) in the voltage range of 2.2–4.5 V (vs Na⁺/Na) using a battery charge/discharge test system (WBCS 3000, WonATech). The electrochemical properties were evaluated by applying a current density of 26 mA g⁻¹, which was similar with to the current density of C/10 (IC = 254.4 mA g⁻¹), as well as its multiples. In the case of the Na-deficient layered-type SIB cathode like P3-NNMVO, more Na⁺ ions can be inserted compared to the pristine state.^[56] Therefore, the theoretical capacity of P3-NNMVO was calculated by setting 1 mol Na content that was the maximum possible intercalation amount in the P3-NNMVO structure, instead of the 0.56 mol Na content in the pristine state. The process for calculating the average voltage in this research was as follows: First, the charge/discharge curves were integrated based on capacity (in mAh g⁻¹) versus voltage (in V) as specified to obtain the energy density (in mWh g⁻¹ (= Wh kg⁻¹)). Then, the energy density was divided by the capacity (in mAh g⁻¹) to calculate the precise values of the average voltage (in V). This method can provide the precise value of the average voltage.

Computational Details: The density functional theory (DFT) calculations were conducted with the projector augmented wave (PAW) method,^[68] which was implemented in Vienna Ab initio Simulation Package (VASP).^[69] For the calculations, supercell shape of $\text{P3-Na}_x[\text{Ni}_{0.1}\text{Mn}_{0.81}\square_{0.09}]\text{O}_2$ was $2\times 2\times 1$ and the k-point grid was $6\times 6\times 3$. The DFT calculations of all $\text{Na}_x[\text{Ni}_{0.1}\text{Mn}_{0.81}\square_{0.09}]\text{O}_2$ compositions ($x = 0, 0.167, 0.333, 0.5, 0.667, 0.833, 1$) were conducted with maximum 10 configurations with the lowest electrostatic energies in each composition. Those configurations were selected through the cluster expansion using CASM^[70] software and comparing Ewald summation. About detailed conditions, the kinetic energy cut-off was set as 500 eV values to characterize the interaction between the ionic cores and the valence e^- , and the ionic relaxation loops were stopped when the norms of all the forces were smaller than 0.03 eV. The exchange-correlation functional was treated using the Perdew-Burke Ernzerhof (PBE), which was one of the generalized gradient approximation (GGA).^[71] The localization of the d-orbital in Mn and Ni ions were represented by GGA+U method with the effective U values of 3.9 eV and 6.0 eV, respectively.^[72] The Heyd–Scuseria–Ernzerhof (HSE06) functional was adopted to calculate projected density of states (pDOS) of Mn, Ni and O ions.^[73] Nudged elastic band (NEB) calculations were performed to determine the activation barrier for Na^+ diffusion using $\text{Na}_0[\text{Ni}_{0.1}\text{Mn}_{0.81}\square_{0.09}]\text{O}_2$ and $\text{Na}_0[\text{Ni}_{0.1}\text{Mn}_{0.9}]\text{O}_2$ structures each. For NEB calculation, five intermediate images were generated between each Na site.^[74] After setting all Na location, each structures were calculated using the NEB algorithm with fixed lattice parameters and free atomic positions.^[75]

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

S.-Y.L. and H.K. contributed equally to this work. This research was supported National R&D Program through the National Research Foundation of Korea (NRF) funded by Ministry of Science and ICT (RS-2024-00408156) of Republic of Korea. Also, this work was supported by internal fund/grant of Electronics and Telecommunications Research Institute (ETRI) (23YB2600, Development of Fundamental Technology for Non-Lithium Resources Based Next Generation Aqueous-Type Multivalence-Metal-Ion Battery).

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

cathode, layered oxide, Na-ion battery, oxygen redox, P3-type

Received: June 3, 2024

Revised: July 20, 2024

Published online: August 3, 2024

- [2] J. B. Goodenough, *Nat. Electron.* **2018**, 1, 204.
- [3] S. J. Davis, N. S. Lewis, M. Shaner, S. Aggarwal, D. Arent, I. L. Azevedo, S. M. Benson, T. Bradley, J. Brouwer, Y. M. Chiang, C. T. M. Clack, A. Cohen, S. Doig, J. Edmonds, P. Fennell, C. B. Field, B. Hannegan, B. M. Hodge, M. I. Hoffert, E. Ingersoll, P. Jaramillo, K. S. Lackner, K. J. Mach, M. Mastrandrea, J. Ogden, P. F. Peterson, D. L. Sanchez, D. Sperling, J. Stagner, J. E. Trancik, et al., *Science* **2018**, 360, eaas9793.
- [4] C. Xu, P. Behrens, P. Gasper, K. Smith, M. Hu, A. Tukker, B. Steubing, *Nat. Commun.* **2023**, 14, 119.
- [5] P. R. Tete, M. M. Gupta, S. S. Joshi, *J. Energy Storage* **2021**, 35, 102255.
- [6] R. Zhang, C. Wang, P. Zou, R. Lin, L. Ma, T. Li, I. hui Hwang, W. Xu, C. Sun, S. Trask, H. L. Xin, *Nat. Energy* **2023**, 8, 695.
- [7] Y. Lee, H. Park, M. kyung Cho, J. Ahn, W. Ko, J. Kang, Y. J. Choi, H. Kim, I. Park, W. H. Ryu, J. Hong, J. Kim, *Adv. Funct. Mater.* **2022**, 32, 1.
- [8] E. M. Bibra, E. Connelly, S. Dhir, M. Drtil, P. Henriot, I. Hwang, J.-B. L. Marois, S. McBain, L. Paoli, J. Teter, Global EV Outlook 2022, IEA, France, Paris **2022**.
- [9] Y. Liang, R. Kleijn, A. Tukker, E. van der Voet, *Renew. Sustain. Energy Rev.* **2022**, 161, 112334.
- [10] W. Ko, J. Kim, J. Kang, H. Park, Y. Lee, J. Ahn, B. Ku, M. Choi, H. Ahn, G. Oh, J. Y. Hwang, J. Kim, *Mater. Today Energy* **2023**, 36, 101356.
- [11] B. Ku, H. Ahn, S. Lee, J. Ahn, M. Choi, J. Kang, H. Park, J. Kim, A. Y. Kim, H. G. Jung, J. K. Yoo, J. Kim, *Energy Storage Mater.* **2023**, 62, 102952.
- [12] X. Wang, Q. Zhang, C. Zhao, H. Li, B. Zhang, G. Zeng, Y. Tang, Z. Huang, I. Hwang, H. Zhang, S. Zhou, Y. Qiu, Y. Xiao, J. Cabana, C. J. Sun, K. Amine, Y. Sun, Q. Wang, G. L. Xu, L. Gu, Y. Qiao, S. G. Sun, *Nat. Energy* **2024**, 9, 184.
- [13] J. Kang, J. Ahn, H. Park, W. Ko, Y. Lee, S. Lee, S. Lee, S. K. Jung, J. Kim, *Adv. Funct. Mater.* **2022**, 32, 1.
- [14] M. Choi, H. Ahn, H. Park, Y. Lee, J. Ahn, B. Ku, J. Kim, W. Ko, J. Kang, J. K. Yoo, D. Kim, J. Kim, *J. Energy Chem.* **2023**, 85, 144.
- [15] T. Chen, B. Ouyang, X. Fan, W. Zhou, W. Liu, K. Liu, *Carbon Energy* **2022**, 4, 170.
- [16] M. Kim, H. Kim, M. Cho, D. Kim, *J. Mater. Chem. A* **2021**, 9, 15179.
- [17] X. Cao, H. Li, Y. Qiao, X. Li, M. Jia, J. Cabana, H. Zhou, *Adv. Energy Mater.* **2020**, 10, 1.
- [18] A. Konarov, J. H. Jo, J. U. Choi, Z. Bakenov, H. Yashiro, J. Kim, S. T. Myung, *Nano Energy* **2019**, 59, 197.
- [19] U. Maitra, R. A. House, J. W. Somerville, N. Tapia-Ruiz, J. G. Lozano, N. Guerrini, R. Hao, K. Luo, L. Jin, M. A. Pérez-Osorio, F. Massel, D. M. Pickup, S. Ramos, X. Lu, D. E. McNally, A. V. Chadwick, F. Giustino, T. Schmitt, L. C. Duda, M. R. Roberts, P. G. Bruce, *Nat. Chem.* **2018**, 10, 288.
- [20] S. M. Kang, D. Kim, K. S. Lee, M. S. Kim, A. Jin, J. H. Park, C. Y. Ahn, T. Y. Jeon, Y. H. Jung, S. H. Yu, J. Mun, Y. E. Sung, *Adv. Sci.* **2020**, 7, 1.
- [21] Q. C. Wang, J. K. Meng, X. Y. Yue, Q. Q. Qiu, Y. Song, X. J. Wu, Z. W. Fu, Y. Y. Xia, Z. Shadike, J. Wu, X. Q. Yang, Y. N. Zhou, *J. Am. Chem. Soc.* **2019**, 141, 840.
- [22] H. Hirsh, Y. Li, J.-H. Cheng, R. Shimizu, M. Zhang, E. Zhao, Y. S. Meng, *J. Electrochem. Soc.* **2021**, 168, 040539.
- [23] J. Liu, C. Didier, M. Sale, N. Sharma, Z. Guo, V. K. Peterson, C. D. Ling, *J. Mater. Chem. A* **2020**, 8, 21151.
- [24] F. Li, R. Liu, J. Liu, H. Li, *Adv. Funct. Mater.* **2023**, 33, 1.
- [25] L. Yang, Z. Liu, S. Liu, M. Han, Q. Zhang, L. Gu, Q. Li, Z. Hu, X. Wang, H.-J. Lin, C.-T. Chen, J.-M. Chen, S.-C. Haw, Z. Wang, L. Chen, *Nano Energy* **2020**, 78, 105172.
- [26] Z. Xiao, W. Zuo, X. Liu, J. Xie, H. He, Y. Xiang, H. Liu, Y. Yang, *ACS Appl. Mater. Interfaces* **2021**, 13, 38305.
- [27] H. Liu, X. Gao, J. Chen, J. Gao, S. Yin, S. Zhang, L. Yang, S. Fang, Y. Mei, X. Xiao, L. Chen, W. Deng, F. Li, G. Zou, H. Hou, X. Ji, *J. Power Sources* **2021**, 508, 230324.

[1] M. Aneke, M. Wang, *Appl. Energy* **2016**, 179, 350.

- [28] Z. Zhu, D. Yu, Y. Yang, C. Su, Y. Huang, Y. Dong, I. Waluyo, B. Wang, A. Hunt, X. Yao, J. Lee, W. Xue, J. Li, *Nat. Energy* **2019**, *4*, 1049.
- [29] J. H. Paterson, O. L. Krivanek, *Ultramicroscopy* **1990**, *32*, 319.
- [30] A. Yang, Y. Wang, F. Yang, D. Wang, Y. Zi, K. L. Tsui, B. Zhang, *J. Power Sources* **2019**, *443*, 227108.
- [31] J. H. Jo, J. U. Choi, A. Konarov, H. Yashiro, S. Yuan, L. Shi, Y. K. Sun, S. T. Myung, *Adv. Funct. Mater.* **2018**, *28*, 1.
- [32] T. Yang, N. Zhang, Y. Lang, K. Sun, *Electrochim. Acta* **2011**, *56*, 4058.
- [33] Y. Yu, D. Ning, Q. Li, A. Franz, L. Zheng, N. Zhang, G. Ren, G. Schumacher, X. Liu, *Energy Storage Mater.* **2021**, *38*, 130.
- [34] X. Rong, J. Liu, E. Hu, Y. Liu, Y. Wang, J. Wu, X. Yu, K. Page, Y. S. Hu, W. Yang, H. Li, X. Q. Yang, L. Chen, X. Huang, *Joule* **2018**, *2*, 125.
- [35] J. H. Jo, H. J. Kim, J. U. Choi, N. Voronina, K. S. Lee, K. Ihm, H. K. Lee, H. D. Lim, H. Kim, H. G. Jung, K. Y. Chung, H. Yashiro, S. T. Myung, *Energy Storage Mater.* **2022**, *46*, 329.
- [36] B. Song, E. Hu, J. Liu, Y. Zhang, X. Q. Yang, J. Nanda, A. Huq, K. Page, *J. Mater. Chem. A* **2019**, *7*, 1491.
- [37] B. Chen, Y. Xin, Y. Wang, X. Ding, C. Jiang, F. Wu, H. Gao, *Energy Storage Mater.* **2024**, *67*, 103252.
- [38] Y. Zhang, M. Wu, J. Ma, G. Wei, Y. Ling, R. Zhang, Y. Huang, *ACS Cent. Sci.* **2020**, *6*, 232.
- [39] Y. Lai, H. Xie, P. Li, B. Li, A. Zhao, L. Luo, Z. Jiang, Y. Fang, S. Chen, X. Ai, D. Xia, Y. Cao, *Adv. Mater.* **2022**, *34*, 1.
- [40] A. Massaro, A. B. Muñoz-García, P. P. Prosini, C. Gerbaldi, M. Pavone, *ACS Energy Lett.* **2021**, *6*, 2470.
- [41] T. Song, C. Wang, C. Lee, *Carbon Neutralization* **2022**, *1*, 68.
- [42] N. Dagar, S. Saxena, H. N. Vasavan, A. K. Das, P. Gami, S. Deswal, P. Kumar, S. Kumar, *Mater. Lett.* **2024**, *369*, 136768.
- [43] E. J. Kim, L. A. Ma, L. C. Duda, D. M. Pickup, A. V. Chadwick, R. Younesi, J. T. S. Irvine, A. Robert Armstrong, *ACS Appl. Energy Mater.* **2020**, *3*, 184.
- [44] M. M. Kalantarian, H. Y. Mashhour, *Electrochim. Acta* **2020**, *343*, 136127.
- [45] H. L. Yu, K. B. Ibrahim, P. W. Chi, Y. H. Su, W. T. Chen, S. C. Tseng, M. T. Tang, C. L. Chen, H. Y. Tang, C. W. Pao, K. H. Chen, M. K. Wu, H. L. Wu, *Adv. Funct. Mater.* **2022**, *32*, 1.
- [46] N. Voronina, M. Y. Shin, H. J. Kim, N. Yaqoob, O. Guillon, S. H. Song, H. Kim, H. D. Lim, H. G. Jung, Y. Kim, H. K. Lee, K. S. Lee, K. Yazawa, K. Gotoh, P. Kaghazchi, S. T. Myung, *Adv. Energy Mater.* **2022**, *12*, 1.
- [47] J. Sun, C. Sheng, X. Cao, P. Wang, P. He, H. Yang, Z. Chang, X. Yue, H. Zhou, *Adv. Funct. Mater.* **2022**, *32*, 2110295.
- [48] C. Zhao, H. Liu, F. Geng, B. Hu, C. Li, *Energy Storage Mater.* **2022**, *48*, 290.
- [49] H. Liu, C. Li, W. Tong, B. Hu, *J. Phys. Chem. Lett.* **2023**, *14*, 2323.
- [50] C. Tian, D. Nordlund, H. L. Xin, Y. Xu, Y. Liu, D. Sokaras, F. Lin, M. M. Doeff, *J. Electrochem. Soc.* **2018**, *165*, A696.
- [51] P. GE, *Solid State Ionics* **1988**, *28*, 1172.
- [52] F. Wang, B. Peng, S. Zeng, L. Zhao, X. Zhang, G. Wan, H. Zhang, G. Zhang, *Adv. Funct. Mater.* **2022**, *32*, 1.
- [53] T. Song, C. Wang, L. Kang, W. Yao, H. Wang, H. Chen, Q. Liu, Y. Lu, Z. Guan, A. Zhu, T. Kang, Y. Tang, C. S. Lee, *Adv. Energy Mater.* **2023**, *13*, 1.
- [54] S. Guo, Y. Sun, J. Yi, K. Zhu, P. Liu, Y. Zhu, G. Zhu, M. Chen, M. Ishida, H. Zhou, *NPG Asia Mater* **2016**, *8*, e266.
- [55] X. L. Li, J. Bao, Z. Shadike, Q. C. Wang, X. Q. Yang, Y. N. Zhou, D. Sun, F. Fang, *Angew. Chemie – Int. Ed.* **2021**, *60*, 22026.
- [56] X. Wu, C. Chen, C. Zhao, H. Liu, B. Hu, J. Li, C. Li, B. Hu, *Small* **2023**, *19*, 1.
- [57] X. Wang, X. Fan, X. Yu, S. Bak, Z. Shadike, I. Waluyo, A. Hunt, S. D. Senanayake, H. Li, L. Chen, C. Wang, R. Xiao, E. Hu, X. Q. Yang, *Adv. Funct. Mater.* **2021**, *31*, 1.
- [58] W. S. Yoon, M. Balasubramanian, K. Y. Chung, X. Q. Yang, J. McBreen, C. P. Grey, D. A. Fischer, *J. Am. Chem. Soc.* **2005**, *127*, 17479.
- [59] R. A. House, L. Jin, U. Maitra, K. Tsuruta, J. W. Somerville, D. P. Förstermann, F. Massel, L. Duda, M. R. Roberts, P. G. Bruce, *Energy Environ. Sci.* **2018**, *11*, 926.
- [60] Q. Li, D. Ning, D. Wong, K. An, Y. Tang, D. Zhou, G. Schuck, Z. Chen, N. Zhang, X. Liu, *Nat. Commun.* **2022**, *13*, 1.
- [61] D. Eum, B. Kim, S. J. Kim, H. Park, J. Wu, S. P. Cho, G. Yoon, M. H. Lee, S. K. Jung, W. Yang, W. M. Seong, K. Ku, O. Tamwattana, S. K. Park, I. Hwang, K. Kang, *Nat. Mater.* **2020**, *19*, 419.
- [62] M. Jia, H. Li, Y. Qiao, L. Wang, X. Cao, J. Cabana, H. Zhou, *ACS Appl. Mater. Interfaces* **2020**, *12*, 38249.
- [63] R. Qiao, Y. De Chuang, S. Yan, W. Yang, *PLoS One* **2012**, *7*, 3.
- [64] W. Yin, J. Alvarado, E. A. Kedzie, B. D. McCloskey, C. Dun, J. J. Urban, Z. Zhuo, W. Yang, M. M. Doeff, *J. Mater. Chem. A* **2023**, *11*, 23048.
- [65] J. Xiao, X. Li, K. Tang, D. Wang, M. Long, H. Gao, W. Chen, C. Liu, H. Liu, G. Wang, *Mater. Chem. Front.* **2021**, *5*, 3735.
- [66] P. Wachulak, T. Fok, A. Bartnik, K. A. Janulewicz, H. Fiedorowicz, *Appl. Phys. B Lasers Opt.* **2020**, *126*, 11.
- [67] S. Kaewmala, N. Kamma, S. Buakeaw, W. Limphirat, J. Nash, S. Srilomsak, P. Limthongkul, N. Meethong, *Sci. Rep.* **2023**, *13*, 4526.
- [68] P. E. Blöchl, *Phys. Rev. B* **1994**, *50*, 17953.
- [69] G. Kresse, J. Furthmüller, *Comput. Mater. Sci.* **1996**, *6*, 15.
- [70] A. Van der Ven, J. C. Thomas, Q. Xu, J. Bhattacharya, *Math. Comput. Simul.* **2010**, *80*, 1393.
- [71] J. P. Perdew, K. Burke, M. Ernzerhof, *Phys. Rev. Lett.* **1996**, *77*, 3865.
- [72] A. Jain, G. Hautier, S. P. Ong, C. J. Moore, C. C. Fischer, K. A. Persson, G. Ceder, *Phys. Rev. B* **2011**, *84*, 045115.
- [73] J. Heyd, G. E. Scuseria, M. Ernzerhof, *J. Chem. Phys.* **2006**, *124*, 219906.
- [74] G. Henkelman, B. P. Uberuaga, H. Jónsson, *J. Chem. Phys.* **2000**, *113*, 9901.
- [75] G. Henkelman, H. Jónsson, *J. Chem. Phys.* **2000**, *113*, 9978.
- [76] D. C. Koningsberger, R. Prins, *X-Ray Absorption: Principles, Applications, Techniques of EXAFS, SEXAFS and XANES*, Wiley-Interscience, United States, NY **1988**.